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Leveraging the power of AI



Team Details

a. **Team name: HELL FIRE CLUB**
b **Team leader name: Payyavula Madhuri**
c. **Problem Statement: AI-Based Crop Recommendation for Farmers**

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Problem Statement :

AI-Based Crop Recommendation for Farmers

Farmers lack a scientific, localized, and profit-oriented decision-support system to select the right crop for their land.

This leads to:

- Crop failures due to unsuitable soil or weather
- Low income because of market price crashes
- Excess production of a single crop
- High dependency on subsidies and loans

Approach to solve that problem:

We propose an AI-based hyper-local crop recommendation platform that provides scientific, profit-aware, and explainable crop advisories to farmers using easily accessible interface

Impact of the Solution

- 📈 Increases farmer income by recommending profitable crops
- 🌾 Reduces crop failure risk
- 💧 Promotes sustainable water usage
- 🧑‍🌾 Improves trust through transparent AI

Limitations in existing solution :

1. Existing solution They do not consider plot-level soil and climate variations.
2. Current advisories focus on yield, not future market prices.
3. Recommendations are given without explanation and also they donot know why that crop was recommended.
4. Language and literacy barriers

How it was different and how to solve :

1. AI-Driven Recommendation Engine
   - Uses machine learning models (Random Forest / XGBoost)
   - Trained on soil, weather, and crop datasets
2. Market-Aware Intelligence
   - Predicts future crop prices using historical mandi data
3. Explainable AI (Trust Layer)
   - Shows how factors like soil pH, rainfall, and demand influenced the decision
4. Climate Risk & Crop Rotation
   - Alerts farmers about drought, flood, or heatwave risks
5. Multi-Channel Accessibility
   - Mobile application
   - WhatsApp chatbot
   - Voice-based advisory in regional languages

List of features in the solution:

- Market-aware crop recommendation (future price prediction)
- Explainable AI – shows why a crop is recommended
- Crop rotation and diversification suggestions
- Climate risk alerts (drought / excess rainfall / heatwave)
- Regional language support
- Voice-based advisory

Google Technologies used in the solution:

Antigravity

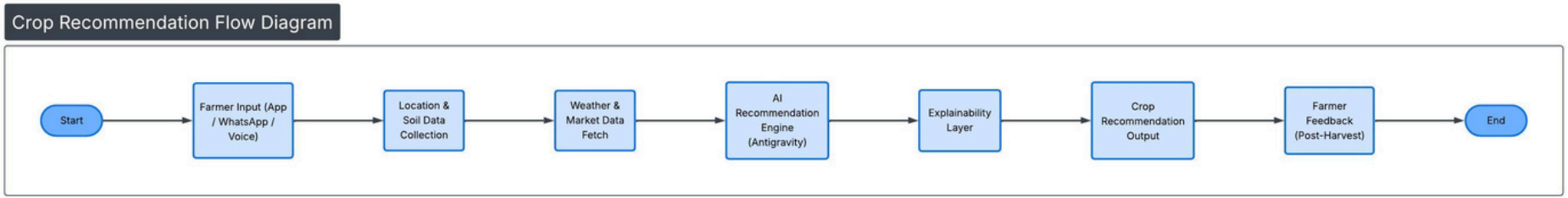
- Used for AI-based crop recommendation modeling
- Handles:
- Soil–crop suitability mapping
- Weather impact analysis
- Yield prediction
- Enables scalable ML workflows

Firebase

- Firebase Authentication – farmer login
- Firestore Database – crop, soil, and recommendation storage
- Firebase Hosting – web dashboard
- Firebase Cloud Functions – API logic

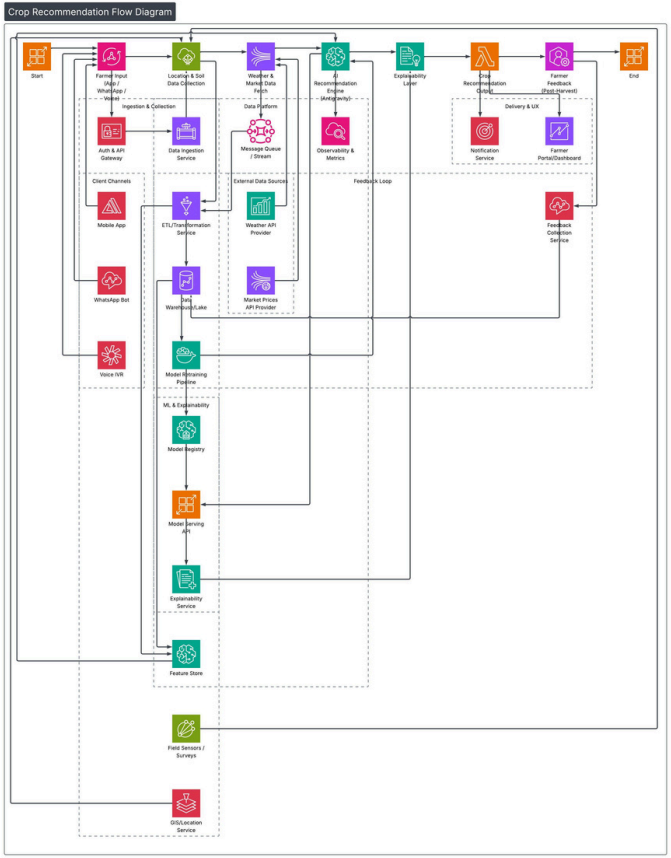
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Process flow diagram :



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Architecture diagram of the proposed solution



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Thank you!